

Talk 7. The Eisenstein Ideal

$N > 7$ prime & $N \neq 13$

Recall: A

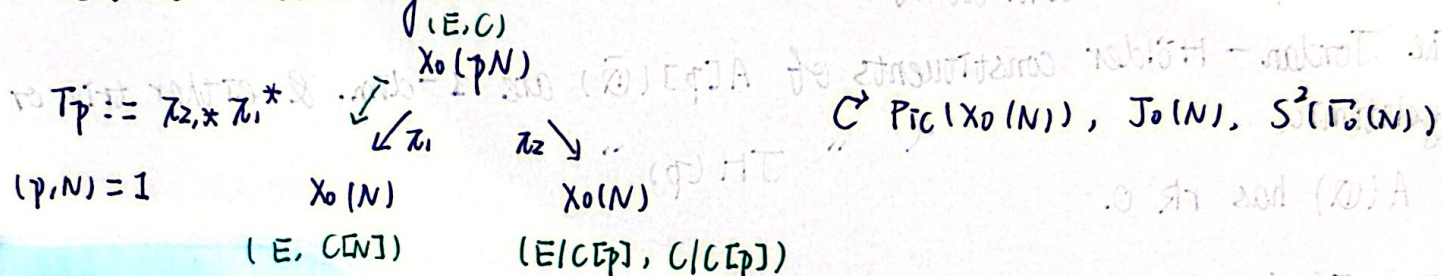
Thm: Suppose $\exists$ an ab. var. $A/\mathbb{Q}$ w/ a map $f: X_0(N) \rightarrow A$ st.

- 1) A has good redn away from N ;
- 2) $f(0) \neq f(\infty)$;
- 3) $A(\mathbb{Q})$ has rk 0.

Then $\nexists$ ell. curve $/\mathbb{Q}$ w/ a rat'l pt. of order N .

Construct A as a quotient of $J_0(N)$.

§ Reminder: Hecke algebras



$X_0(pN)$ has four cusps corresp. to $1, 1/p, 1/N, 1/pN \in \Gamma_0(N)$ -orbits of $\mathbb{P}^1(\mathbb{Q})$

$(0,0) \quad (0,\infty) \quad (\infty,0) \quad (\infty,\infty)$

$\tau_i(x,y) = x$ for $(x,y) \in \curvearrowright$. τ_1 : ramn. index 1 at (x,∞) & p at $(x,0)$
 τ_2 : opposite

$T_p([0]) = (p+1)[0], T_p([\infty]) = (p+1)[\infty]$ in $\text{Pic}(X_0(N))$.

$\Pi :=$ subring of $\text{End}(J_0(N))$ gen. by $\{T_p: p \nmid N\}$.

Fact: 1) Π is commutative & fin. free $\mathbb{Z}$

2) $\Pi \otimes \mathbb{Q} \xrightarrow{\sim} \Pi_f \otimes K_f$, f runs through Galois conj. cls. of normalized eigenforms in $S^2(\Gamma_0(N))$

$\Rightarrow \Pi \hookrightarrow \Pi_f \otimes K_f$

Let $P_f := \ker(\Pi \rightarrow \mathbb{Q}_{K_f})$, then $\exists$ isogeny $J_0(N) \rightarrow \bigoplus_f A_f$, where $A_f := J_0(N)/P_f J_0(N)$ is an ab. var. of genus $g_f = [K_f: \mathbb{Q}]$.

3) $A \otimes \mathbb{C} \rightarrow A_f(\bar{\mathbb{Q}})$ & $V_p(A_f) := \varprojlim_n A_f[\mathbb{P}^n](\bar{\mathbb{Q}})$ is a $2g_f$ -dim. p -adic Galois rep.
 & $V_p(A_f)$ is free $\bigvee K_f$ of rk ≥ 1 $K_f \otimes_{\mathbb{Q}} \bar{\mathbb{Q}}_p = \prod_{\lambda|p} K_\lambda$
 $\Rightarrow V_p(A_f) \simeq \prod_{\lambda|p} V_{f,\lambda}$ w/ $V_{f,\lambda}$ 2 -dim. K_λ -rep.

§ The Eisenstein Ideal

Recall:

⚠ Assumption: All the f 's are rat'l.

Thm^B: $A/\mathbb{Q}$ ab. var., $p \neq N$ prime. If $N > 7, N \neq 13$

- 1) A has good redn away from N
 - 2) A has complete toric redn. at N
 - 3) The Jordan-Hölder constituents of $A[\mathbb{P}](\bar{\mathbb{Q}})$ are 1-dim. & either triv. or cyclotomic.
- " JH(p) "

Then $A(\mathbb{Q})$ has rk 0.

Lem: If f is rat'l ($K_f = \mathbb{Q}$, A_f ell. curve), then A_f satisfying JH(p)

$\Leftrightarrow a_\ell(f) \equiv \ell+1 \pmod{p}, \forall \ell \neq N$.

Pf: " $\Rightarrow$ " $A_f[\mathbb{P}]^{\text{ss}} \simeq \mathbb{Q} \oplus \chi_p \Rightarrow \checkmark$

" $\Leftarrow$ " Chebotarev + Brauer - Nesbitt. Fact: $\mathbb{P}/\mathbb{Q} \simeq \mathbb{Z}/p\mathbb{Z}$ when $\square$.

Def: The Eisenstein Ideal $\mathfrak{a} \subset \mathbb{T} := \langle p, \frac{p}{\ell} - (\ell+1), \forall \ell \neq N \rangle$.

Lem: If f is rat'l, $p \neq N$, $\Leftrightarrow$, then A_f satisfies JH(p) $\Leftrightarrow A_f \subset \mathfrak{a} \Leftrightarrow$
 the image of a in $\mathbb{T}/p\mathbb{T}$ is not unit.

Pf: A_f has JH(p) $\Leftrightarrow p \mid a_\ell(f) - (\ell+1) \Leftrightarrow \text{Im}(a) \subset \mathbb{T}/p\mathbb{T}$ is not unit.

$\Leftarrow A_f \subset \mathfrak{a}$

" $\Rightarrow$ " $A_f \not\subset \mathfrak{a} \Rightarrow A_f + \mathfrak{a} = (1) \Rightarrow$ the image of a is unit.

Rmk: The Eisenstein ideal detects congruences between Eisenstein series & cusp forms. □

Lem: $[0] - [\infty] \in J_0(N)(\mathbb{Q})$ is a non-trivial torsion pt of order $(N-1)$.

Pf: ① $[0] - [\infty] \neq 0$, i.e., $\nexists f: X_0(N) \rightarrow \mathbb{P}^1$ w/ $\deg(f) = [0] - [\infty]$. ✓

since $g(X_0(N)) > 0$ $\Rightarrow \deg f = 1$.

② $[0] - [\infty]$ is torsion: consider $\Delta(\mathbb{Z})/\Delta(N\mathbb{Z})$ mod fcn. of level $T_0(N)$. It has divisor $(N-1)([0] - [\infty])$. $\square$

Rmk: Write $n := \text{ord}([0] - [\infty])$, Ogg proved that $n = \frac{N-1}{\gcd(N-1, 12)}$. When $p|n$, $J_0(N)(\mathbb{Q})$ has a p -torsion pt, and so $J_0(N)[p]$ has a copy of the triv. rep. in it. $\Rightarrow \exists A_f$ has JH(p).

Now, if all f 's are rat'l, $J_0(N) \sim \prod_f A_f$ and we can simply pick out the A_p 's that satisfy JH(p): Fix $p|n$ & take

$$A := J_0(N) / I J_0(N) \quad \text{w/} \quad I := \bigcap_{\substack{p|f \text{ c.a.} \\ \text{min.}}} p f = \bigcap_{\substack{p|f \text{ c.a.} \\ \text{min.}}} p f$$

$$\sim \prod_{p|f \text{ c.a.}} A_f.$$

In general, we still take the same A , but now there is no guarantee that A has JH(p). (e.g. some $V_{f,\lambda}$ satisfies JH(p) but some others don't). We'll adapt the proof of Thm B to A .

§ Quotients of $J_0(N)$.

Commutative alg. facts: w/ notations as above,

1) $I = \bigcap_n a^n = \{x : \exists a \in a, \text{ s.t. } ax = x\}.$

2) $Ia = 0$

Lem: $J_0(N)[a^\infty] \xrightarrow{\sim} A[a^\infty]$

Pf: Write $X = J_0(N)[p^\infty]$, $Y = A[p^\infty]$. Then $\exists$ surj. of $\pi_p := \varprojlim_n \pi/p^n \pi$ -modules $X \twoheadrightarrow Y$ w/ kernel IX , i.e., $\exists$ SES

$$0 \rightarrow IX \rightarrow X \rightarrow Y \rightarrow 0.$$

Localization at $a \rightsquigarrow X_a \simeq Y_a.$ $\square$

Prop: $[0] \neq [\infty]$ in A .

③

$$Q \in J_0(N)[a^\infty] = A[a^\infty]. \quad \text{If } \psi \in \text{Hom}(X) \rightarrow \mathbb{A}^1 \text{ is a point, } \psi^*(a) = [0] \in \mathbb{A}^1 \quad \square.$$

Next time: $A[p^n]_a$ is adm. Feb 27

2) $\hat{\pi}_A \otimes_{\pi} A(\mathbb{R})$ is fin. $\Leftarrow$ as in Jed's talk. : motivat. of $[\infty] = [0]$ $\oplus$

3) $A(\mathbb{R})$ is fin. & commutative alg.

$$A := \bigcup_{i \in I} A_i \quad B := \bigcup_{j \in J} B_j \quad \text{then} \quad A \cap B = \bigcup_{i \in I, j \in J} A_i \cap B_j$$

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8 Comments to (iv).

$$\int_{\Sigma} \kappa = \kappa_0 \quad \text{for } \kappa \in \Sigma : \kappa^2 = \kappa_0 \quad \text{and } \kappa = \kappa_0 \quad \text{for } \kappa \in \Sigma \quad (1)$$

$$[{}^{\infty}a]A \simeq [{}^{\infty}a]A \otimes \mathbb{Z}[\frac{1}{p}]$$

$$X \rightarrow Y \text{ w/ kernel } I_X \text{ is } \mathbb{Z}_2 \text{ E 2E2}$$
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ is not chiral

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